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3 November 1986
SUPERSEDING
MIL-D-80076B
10 February 1972

MILITARY SPECIFICATION

DRILLING MACHINES, UPRIGHT, BOX COLUMN, FLOOR, PLAIN TABLE, HAND FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, floor mounted, box column, sliding head, vertical, manual feed, single spindle, and gang drilling machines, with accessories as described herein.

1.2 Classification. The drilling machines shall be of the following styles and sizes. The style and size to be furnished shall be as specified (see 6.2.1).

Style A - One spindle
Style B - Two spindles
Style C - Three spindles
Style D - Four spindles
Style E - Five spindles
Style F - Six spindles

Size	Drill capacity (inches)	Style A Swing (inches)	Styles B thru F Overhang (inches)
12	5/16	12	6
14	3/8	14	7
16	1/2	16	8
18	3/4	18	9
20	1	20	10
24	1-1/4	24	12

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards (OSHA) (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified,

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the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 79 - Electrical Standard for Industrial Machinery.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B5.10 - Machine Tapers.

ANSI B11.8 - Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The drilling machine shall be new and one of the manufacturer's current models capable of operations in accordance with the requirements herein. The drilling machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the drilling machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Reclaimed materials. The drilling machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the drilling machine.

3.2.3 Energy efficiency. The drilling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the

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guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 19 that are applicable to the machine itself. In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.6 Mercury restriction. The drilling machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the drilling machine.

3.2.8 Environmental protection. The drilling machine shall be so designed and constructed that under the operating, service, transportation, and storage conditions described herein, the drilling machine shall not emit materials hazardous to the ecological system.

3.2.9 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent loosening. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

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3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the drilling machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the drilling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.010 inch. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduation that can be read from the operator's normal position.

3.3.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or ISO 54 for the metric (SI) system. The gears shall be proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell G-45.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist primarily of a base, column, drill head assembly, worktable and driving mechanism.

3.4.1 Base. The base shall be a rib-braced casting of iron or iron alloy, or a steel weldment, and shall provide a foundation for the entire drilling machine. Means shall be provided for fastening the machine to the floor. When a coolant system is required under 3.4.12, the sump or reservoir may be cast as an integral part of the base or furnished as a separate external unit. One or more screw jacks for table support and height adjustment shall be mounted on the base. Gang drill shall have a common base arranged to accept a common table and shall have the number of spindles and all of their related components for the style machine required.

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3.4.2 Column. The column shall be a casting or iron or iron alloy, or a steel weldment, reinforced to provide support and minimize deflection. The column shall be of box construction and shall be arranged to receive and carry the drill head assembly. Guideways shall be provided for accepting the sliding head and worktable. Vertical adjustment of both the sliding head and worktable shall be provided with means for locking. The column support surfaces shall be machined to accurately fit those of the base. Columns on gang drills shall be duplicates of single-spindle machines. When specified (see 6.2.1), gang drills shall be furnished with an adjusting type column. The adjusting mechanism shall allow the operator to change the center-to-center distance of the spindles on the machine base. Means for clamping after adjustment shall be provided.

3.4.3 Worktable. The worktable shall be a casting of iron or iron alloy, reinforced for support and deflection. Table height adjustment on the column shall be by a screw jack arrangement, with means for clamping after positioning. Worktables shall be rectangular with trough, having provisions for draining. When specified (see 6.2.1), a replacement type-wear plate, accurately fitted to the worktable top shall be provided. Work surfaces, of either the table or wear plate, shall be machined to not rougher than 75 microinch and be flat within 0.002 inch. When specified (see 6.2.1), T-slots shall be provided for either the worktable or wear plate, as applicable. The number of T-slots and their configuration shall be as specified. Machined T-slots shall be in accordance with ANSI B5.1.

3.4.4 Drillhead assembly. The drillhead assembly shall consist of the drillhead, quill and spindle assembly, power transmission and sliding head, and when required by design, the upper portion of the column. Means for vertical travel of both the spindle and the sliding head shall be incorporated in the assembly design. The assembly shall be operable as a unit and be interchangeable among other machines of the same size and model.

3.4.5 Drillhead. The drillhead shall be designed to house the power transmission. When the drillhead is not an integral part of the column, it shall be held in position by clamps or other positive means.

3.4.6 Quill and spindle assembly. The quill shall be machined of heat-treated alloy steel, or cast steel. The quill shall house the spindle and be fitted to the sliding head. Manual drill feeds shall be accomplished by means of a hand-feed lever through a rack and pinion. A spring assembly or counterweight apparatus shall retract the quill and spindle assembly when it is released. The spring assembly shall be adjustable for tension. Quill guides shall be provided to support and maintain quill alignment through the full travel of the quill. The spindle shall be machined from heat-treated alloy steel and shall be mounted in the quill on ball or roller bearings. Unless otherwise specified (see 6.2.1), the driven end of the spindle shall be multi-splined. Unless otherwise specified, the spindle shall have a tapered socket hose. Taper shall be manufacturers' standard but shall conform to ANSI B5.10. When specified (see 6.2.1), one of the following spindle nose configurations shall be provided.

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a. Shank type for drill chuck.

b. Threaded end for drill chuck.

c. Solid-male tapered type.

3.4.7 Power transmission assembly. The drilling machine shall be powered by an electric motor. The drive shall be through a belt and step pulley arrangement or belt and gears or all gear drive or direct motor spindle drive. Means for adjusting belt tension shall be provided when a belt-type drive is used. When specified (see 6.2.1), power feed shall be provided. The number of feeds and their feed range shall be as specified.

3.4.8 Sliding head. The sliding head shall be mounted on the column, and shall serve as a support for the quill and spindle. The head shall be vertically adjustable. The slides of the head shall accurately fit the ways of the column and shall be provided with means for clamping after adjustment.

3.4.9 Depth stop. Each spindle of each machine shall be provided with a depth stop equipped with a positive locking device. When specified (see 6.2.1), the depth stop shall be provided with an indicating scale marked in increments of not more than 1/16 of an inch.

3.4.10 Worklight. Unless otherwise specified (see 6.2.1), a worklight shall be provided for each spindle. The lamp shall have a protective shield, an on-off switch, and shall be attached to the machine in such a manner as to allow the light to be swung over the tool and workpiece.

3.4.11 Drill chucks. Drill chucks and arbors, when required to be furnished, shall be balanced, high-grade steel, three-jawed, key operated and shall accurately fit the spindle nose. A key shall be provided which can be permanently attached to the machine.

3.4.12 Coolant system. When specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump or reservoir, a power-driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full-flow cooling. The coolant return system shall include a baffle or a strainer which is easily removable for cleaning. The reservoir shall be equipped with means for draining and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow. Means for returning coolant from work area to reservoir shall be provided. The reservoir shall have means for determining fluid level.

3.4.13 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to the requirements of this paragraph and ANSI/NFPA 79. Each drilling machine shall draw all of its electrical power from a single 115/230-volt, single-phase, 60-hertz (Hz) circuit or when specified, (see 6.2.1) machines with motors over 1 hp, 230/460-volt, 3-phase, 60-Hz circuit. Each drilling machine shall have a fused safety

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disconnect switch or circuit breaker. The machine shall be initally wired for operation on 230 volts. An identified terminal for grounding the drilling machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.4.13.1 Motors. Each motor shall be of the induction-type, wired for operation on the voltage specified for the machine and shall be rated for continuous or intermittent duty. Intermittent duty motors shall have not less than 60-minute rating. Each motor shall have sealed, permanently lubricated ball bearings. Unless otherwise specified (see 6.2.1), each motor shall be single speed and totally enclosed. Motor characteristics shall conform to ANSI NEMA MG-1.

3.4.13.2 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.4.13.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary nominal voltage of 120 volts.

3.5 Sizes and capacities. The sizes and capacities of the machine shall meet the requirements in table I.

3.6 Operation. Each machine shall be operated at no load for not less than 30 minutes in accordance with 4.7.1.

3.7 Performance. Each machine shall be capable of drilling 15 through holes, of the rated drill size, in accordance with 4.7.3.

3.8 Alignment tolerance. Each drilling machine shall meet the alignment tolerances specified in table II.

3.9 Standard equipment. Unless otherwise specified (see 6.2.1), the following shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One drill chuck for each spindle, 5/16-inch capacity, for tapered socket spindles only.

c. One extra set of motor driven belts when applicable.

3.10 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1)

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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3.11.1 Lubrication plate. Unless otherwise specified (see 6.2.1). A corrosion-resistant lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of application
Service interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to drilling machine and contain the information listed below. If the drilling machine is a special model, the model designation shall include the model of the basic standard drilling machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviation, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.12 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the drilling machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. Failure of the item to pass the examination, the tests, or the inspections shall be cause for rejection.

4.6 Examination. The drilling machine shall be examined to determine compliance with all requirements of this specification including compliance with all referenced standards.

4.7 Tests.

4.7.1 Operational test. The drilling machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

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4.7.2 Alignment tolerance test. Each drilling machine shall be tested to determine that it complies with the alignment tolerance requirements in table II.

4.7.3 Performance tests. Each spindle of each machine shall be used to drill 15 through holes, of the maximum rated size for the machine, in 1-inch thick mild steel. Drilling time shall be not more than 20 minutes total elapsed time (center drilling time not included). There shall be no evidence of overloading, overheating, excessive vibration or permanent deformation of any parts of the machine during or after drilling operations.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When required, level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058 as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The drilling machines covered by this specification are intended for use in production shops where wide varieties of drilling operations are performed.

6.2 Ordering data. Purchasers should exercise any desired option offered herein, and procurement documents should specify the following:

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Style and size required (see 1.2).

c. First article, if required (see 3.1).

d. If drilling machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Additional safety and health requirements, if required (see 3.2.4).

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f. Painting, if different (see 3.3.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

h. Adjusting type column, if required (see 3.4.2).

i. Wear plate, if required (see 3.4.3).

j. Specify the number and configuration of T-slots, if required (see 3.4.3).

k. Specify when drive end of spindle is not required to be multi-spindle (see 3.4.6).

l. Specify type spindle nose configuration required for drill chuck, if different (see 3.4.6).

m. Power feed if required, and feeds and range (see 3.4.7).

n. Depth stop marked indicator, if required (see 3.4.9).

o. Specify if a worklight is not required (see 3.4.10).

p. Specify if coolant system is not required (see 3.4.12).

q. Electrical requirements, if different (see 3.4.13).

r. Number of speeds and type of enclosures for motors, if different (see 3.4.13.1).

s. Standard equipment, if different (see 3.9).

t. Specify and fully describe optional equipment, if required (see 3.10).

u. Lubrication chart or plate, if different (see 3.11.1).

v. Nameplate, if different (see 3.11.2).

w. Inspection conditions, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (see 4.5).

z. If preservation, packing, and marking in accordance with MIL-M-18058 are required, specify level required (see 5.1).

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6.2.2 Contract data requirements. Required technical data (operating manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing:

Box column
Drilling machine.
Floor mounted
Manual feed
Single spindle and gang drilling
Sliding head
Vertical

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Sizes and capacities.

Characteristics	Size 12	Size 14	Size 16	Size 18	Size 20	Size 24
Capacity (in mild steel), inches	5/16	3/8	1/2	3/4	1	1-1/4
Centerline of spindle to column face, inches	6	7	8	9	10	12
Swing, inches, Style A	12	14	16	18	20	24
Overhang, inches, Styles B thru F	6	7	8	9	10	12
Sliding head adjustment, inches	3-1/2	3-1/2	4	5	5	6
Spindle traverse, inches	3	3	4	5	5	5-3/4
Spindle nose to table, maximum inches	16	20	21	25	25	25
Spindle speeds, number	Infinite	Infinite	Infinite	Infinite	Infinite	Infinite
Spindle speed range, revolutions per minute	625 to 5000	625 to 5000	500 to 4000	400 to 3000	200 to 2000	150 to 1500
Spindle diameter, (at drive end), inches	1/2	19/32	3/4	7/8	1	1-1/8
Quill diameter, inches	1-3/4	1-3/4	1-7/8	1-7/8	2	2-3/8
Distance floor to table, range, inches	25 to 36	25 to 36	25 to 36	25 to 36	25 to 36	25 to 36
Motor horsepower, minimum	1/2	1/2	1	2	3	5

NOTE: Machine sizes and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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TABLE II. Alignment and accuracy tolerance.

Characteristics	Maximum allowable deviation
Working surface of worktable, flat	.0016"/12"
Column guideways square with working surface of the worktable, in longitudinal and traverse plates	.003"/12"
Spindle taper, runout 1 inch from nose	.001" TIR
Spindle taper, runout 6 inches from nose	.002" TIR
Spindle square with working surface of the worktable, turn around method with 6-inch radius, in longitudinal and transverse planes	.003"/12"

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activity:

Preparing activity:
DLA - IP

Project (3413-0139)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-80076C	Drilling Machines, Upright, Box Column, Floor Plain Table, Hand Feed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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